Taghleef Industries

PACKAGING SOLUTIONS WORLDWIDE

# *Ti* Films for IML

# AGENDA

- Label market overview
- *Ti* label market
- *Ti* Product portfolio LABELS
- *Ti* Product portfolio IML films
- New developments

# World market Shares by Labelling Technology 2014

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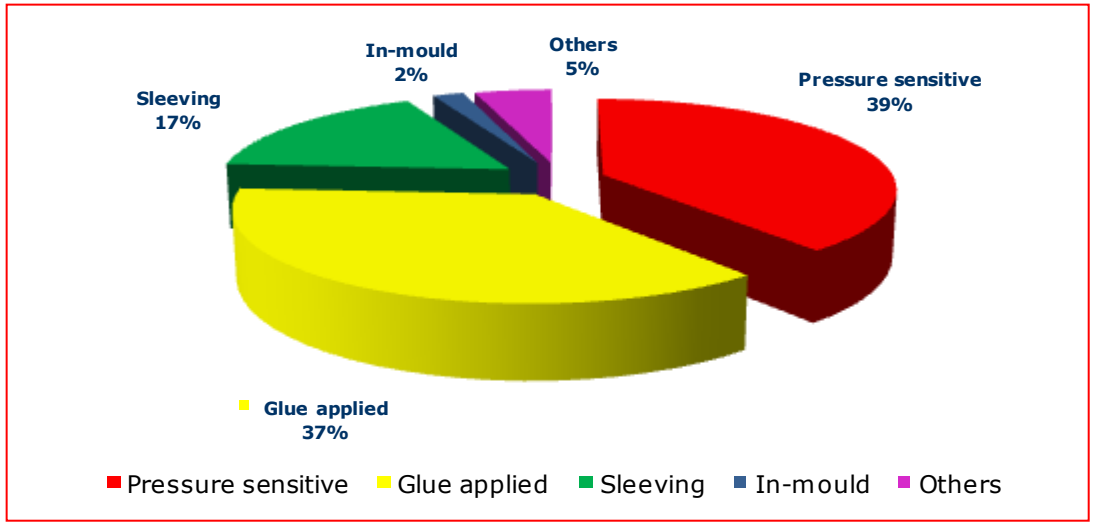
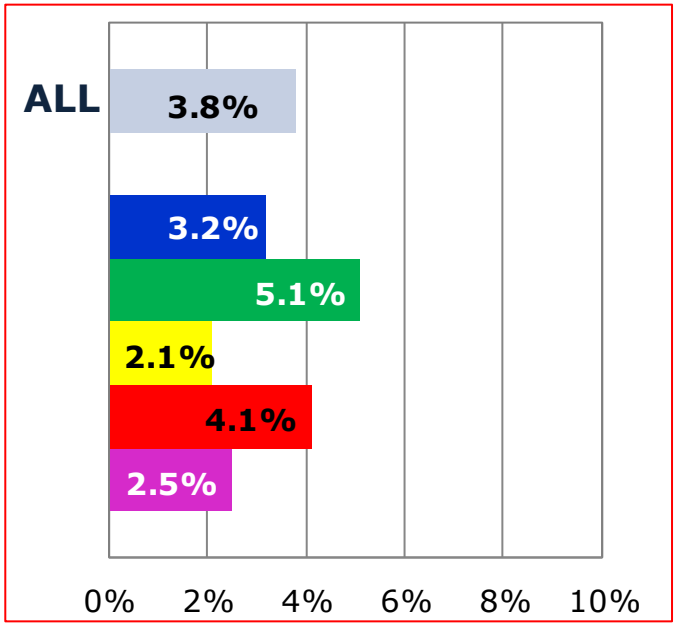
Total market volume: **56 Billion sqm**

Europe : **14 Billion sqm** (25%)

## World market Shares by Technology




## Growth Rates by Technology




Source: AWA

# *Ti* : A wide product range for labels

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- In-Mould
- Self-Adhesive
- WAL Reel-fed
- WAL Cut & Stack
- WAL Shrink - VISION®

NEW to EU

# *Ti* Product Portfolio LABELS

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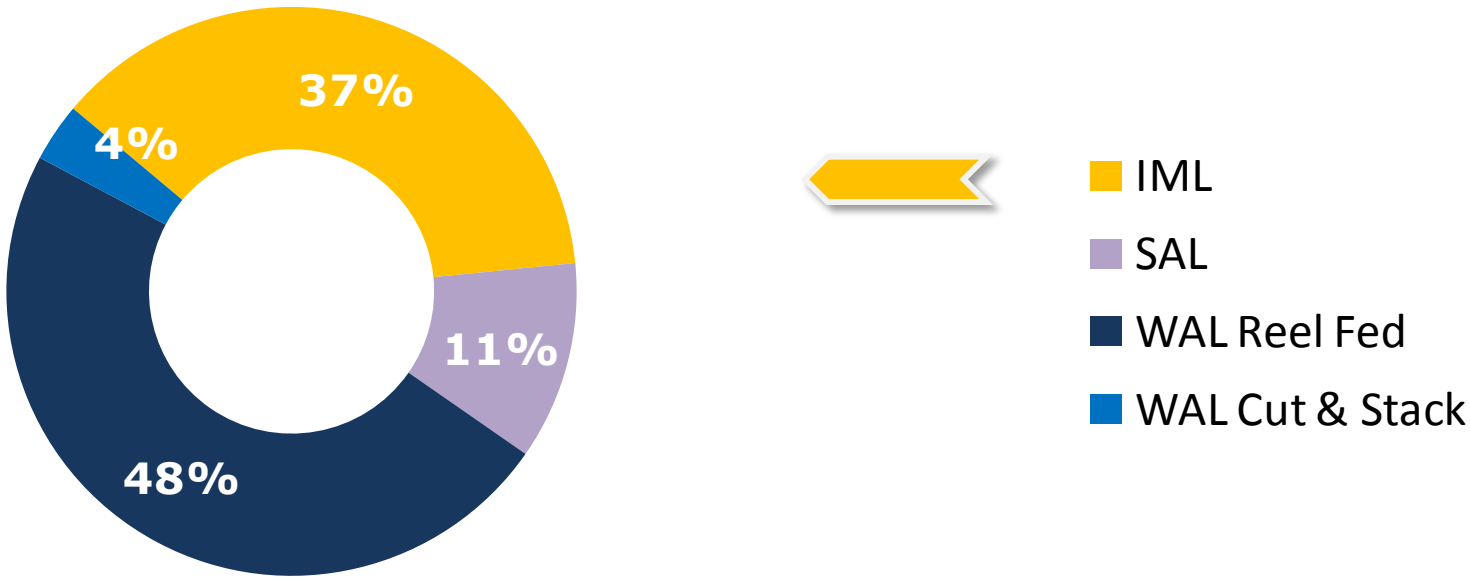
| | IN MOULD LABEL | SELF ADHESIVE LABEL | WRAP AROUND LABEL - Reel Fed | WRAP AROUND LABEL - Cut&Stack | WRAP AROUND LABEL - Shrink |
|---|---|---|---|---|---|
| Transparent | LIT | LSA, LNS, LSR | LTS, LTN | LCT | |
| Matt | | | LUS | | |
| Solid White | LIW | LSW | | | |
| White voided | LIM, LIU, LIX, LIG, LIH, LFE | LSE | LRD, LGL, LXI | LCS | U002 |
| Metallised | | LZP | LZR, LZL | | |

NEW to EU

# *Ti* Europe Volumes 2015
## by Labelling technology



# IML market

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## Regions

- mature market : Europe with major market share
- medium growth rates : North America and Middle East,
- high growth rates : Eastern Europe, South America and Asia,

# IML segments

| Segment |
| --- |
| IML |
| SAL |
| WAL Reel Fed |
| WAL C&S |

- **I – IML**: **I**njection, In Mould Labeling
- **T – IML**: **T**hermoforming, In Mould Labeling

- **B – IML**: Blow mold, In Mould Labeling
- **IMD**: In Mold Decoration *(durable goods)*

# IML applications

- Food (margarine, butter, ice-cream, yoghurt, spreadable cheese)

- Paint containers
- Household
- Garden furniture
- Cosmetics (tubes)

# New markets

## New markets

- Pet food
- ready made meals
- snacks
- confectionery

## New market requirements?

# *Ti* Product Portfolio IML

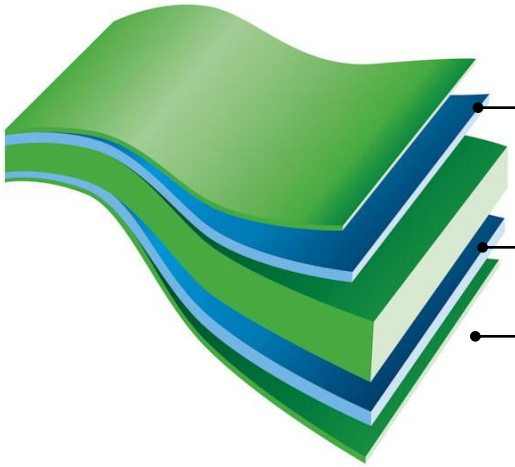
| | I - IML | T- IML |
|---|---|---|
| Transparent | LIT | |
| Solid White | LIW | |
| White voided | LIM, LIG, LIH, LIX New | LFE |
| Metallised | | |

What is your criteria?

# Criteria : aesthetics

- no label look (transparent)
- white / white voided – different finishes

# Transparent : LIT



*Treated layer for printing,* ***NO LABEL LOOK***

*Transparent OPP Core*

*Treated layer for injection moulding*

## LIT55

NEW

**SUPERIOR CLARITY**

## D556 55

# Ultra clear : D556 55 mic

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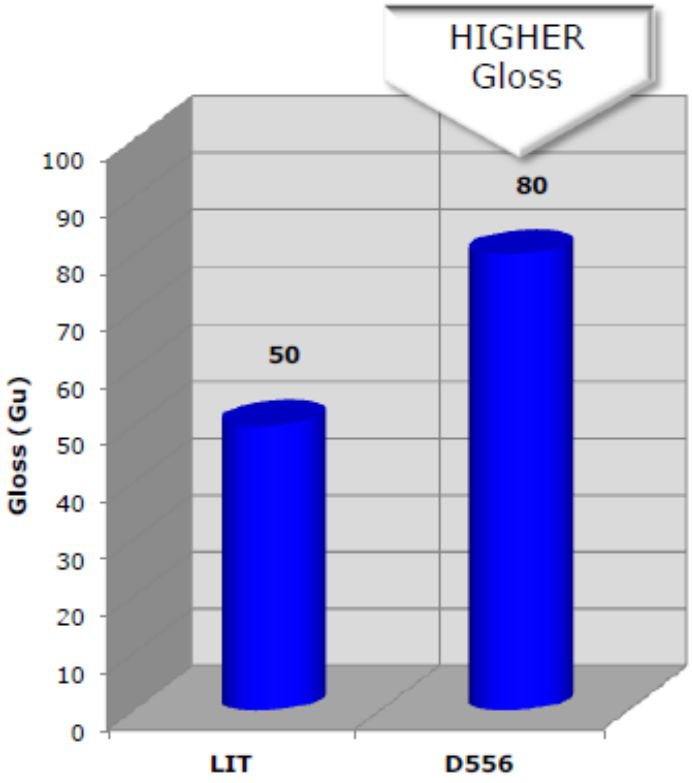
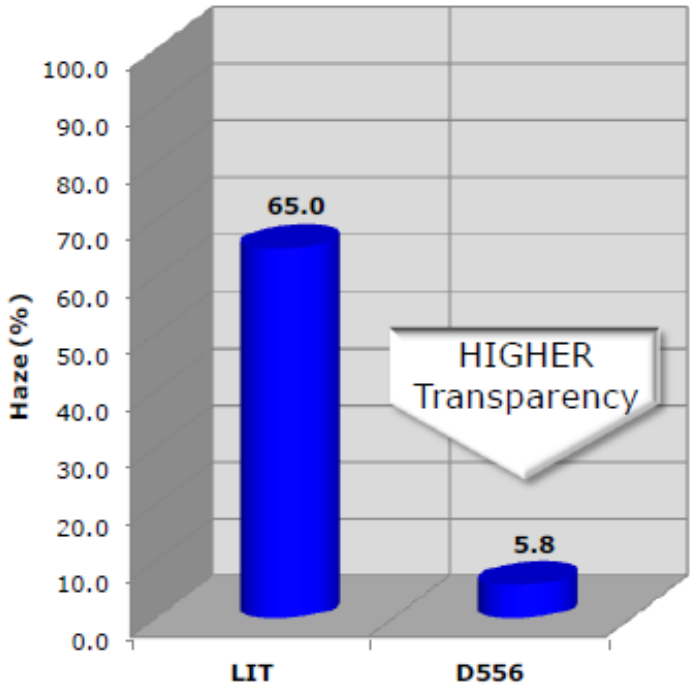
## D556 *NO-Label Look*



# White films : different finishes

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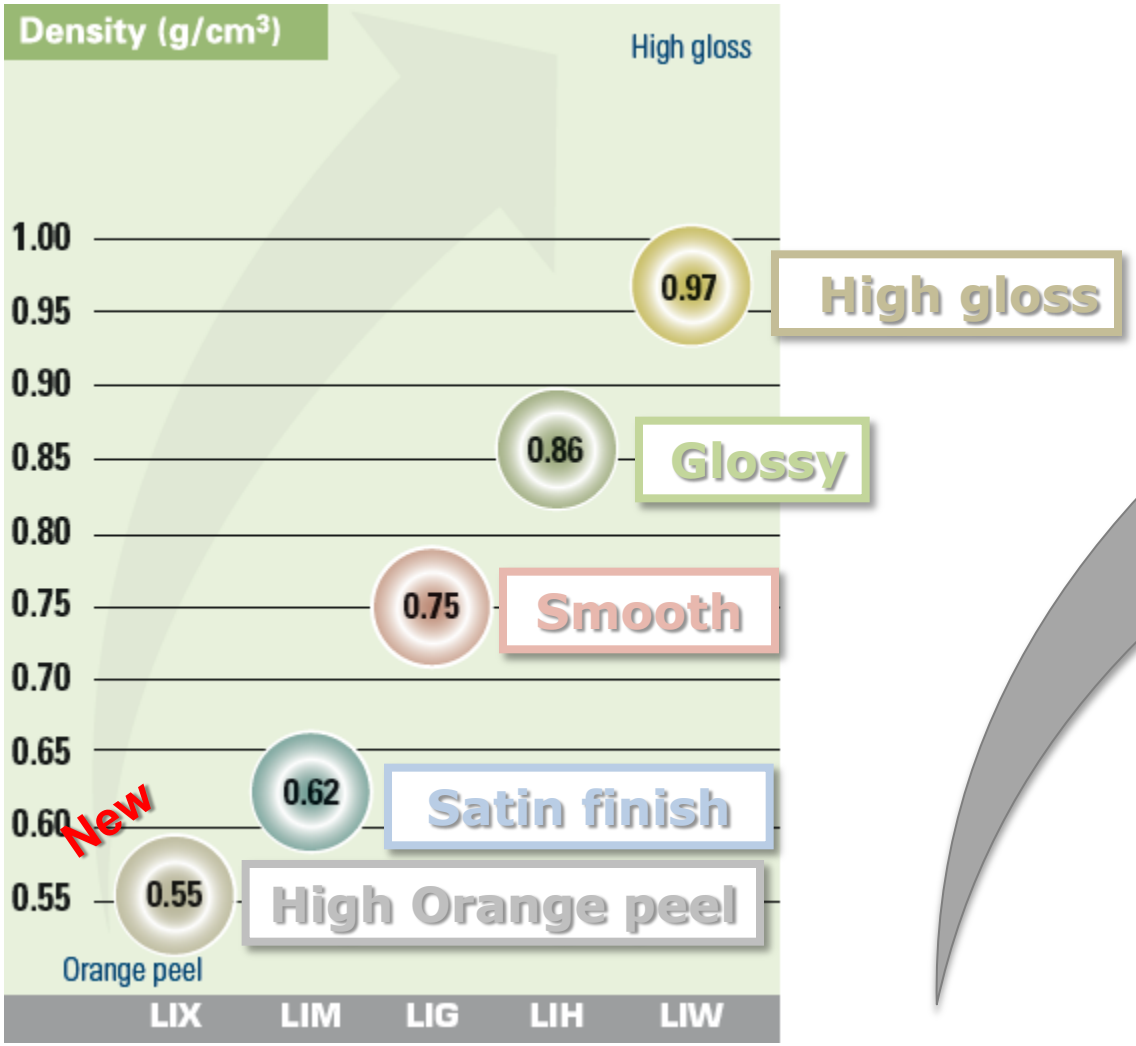


POST MOULD FINISH

# White voided : LIM

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## LIM

- **Satin finish (orange peel)**
- High yield
- *Small / medium* size containers
- Thin / thick wall applications

| Nominal Thickness (µm) | Unit Weight (g/m²) | Yield (m²/kg) | After injection moulding |
|---|---|---|---|
| 45 | 27.9 | 35.8 | Orange peel |
| 55 | 34.1 | 29.3 | |
| 60 | 37.2 | 26.9 | |
| 65 | 40.3 | 24.8 | |
| Density (g/cm³): 0.62 | | | |

# White voided : LIX

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## LIX

New

- **High orange peel**
- Ultra High yield d=0.55 g/cm$^3$
- 50 µm: good converting performance
- Different thickness ⇔ Different container sizes

| Nominal Thickness (µm) | Unit Weight (g/m²) | Yield (m²/kg) | After injection moulding |
|---|---|---|---|
| 50 | 27.5 | 36.4 | High Orange peel |
| 60 | 33.0 | 30.3 | |
| 70 | 38.5 | 26.0 | |
| | | | |
| Density (g/cm³): 0.55 | | | |

# White voided : LIG

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## LIG

- **Smooth finish**
- Optimum stifness
- *Medium size* containers

| Nominal Thickness (µm) | Unit Weight (g/m²) | Yield (m²/kg) | After injection moulding |
|---|---|---|---|
| 70 | 52.5 | 19.0 | Smooth |
| | | | |
| | | | |
| | | | |
| Density (g/cm³): 0.75 | | | |

# White voided : LIH

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## LIH

- **Glossy** post moulding finish
- Enhanced stiffness
- *Medium / Large* containers

| Nominal Thickness (µm) | Unit Weight (g/m²) | Yield (m²/kg) | After injection moulding |
|---|---|---|---|
| 70 | 60.2 | 16.6 | Glossy |
| | | | |
| | | | |
| | | | |
| Density (g/cm³): 0.86 | | | |

New

- Smooth Thin wall : new thickness 50 µm

# Solid white :LIW

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## LIW

- **High Gloss** post moulding finish
- good stiffness

| Nominal Thickness (µm) | Unit Weight (g/m²) | Yield (m²/kg) | After injection moulding |
|---|---|---|---|
| 50 | 48.3 | 20.7 | High gloss |
| | | | |
| | | | |
| | | | |
| Density (g/cm³): 0.97 | | | |

# Post moulding finishes

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| New | New | | | | | New | |
|---|---|---|---|---|---|---|---|
| LIU | LIX | LIM | LIG | LIH | LIT | LTZ | LIW |
| 0.55 | 0.55 | 0.62 | 0.75 | 0.86 | 0.91 | 0.91 | 0.96 |

**Density (g/cm³)**

**"Orange peel"** | **"Satin"** | **"Smooth"** | **Glossy**

# Criteria: stifness

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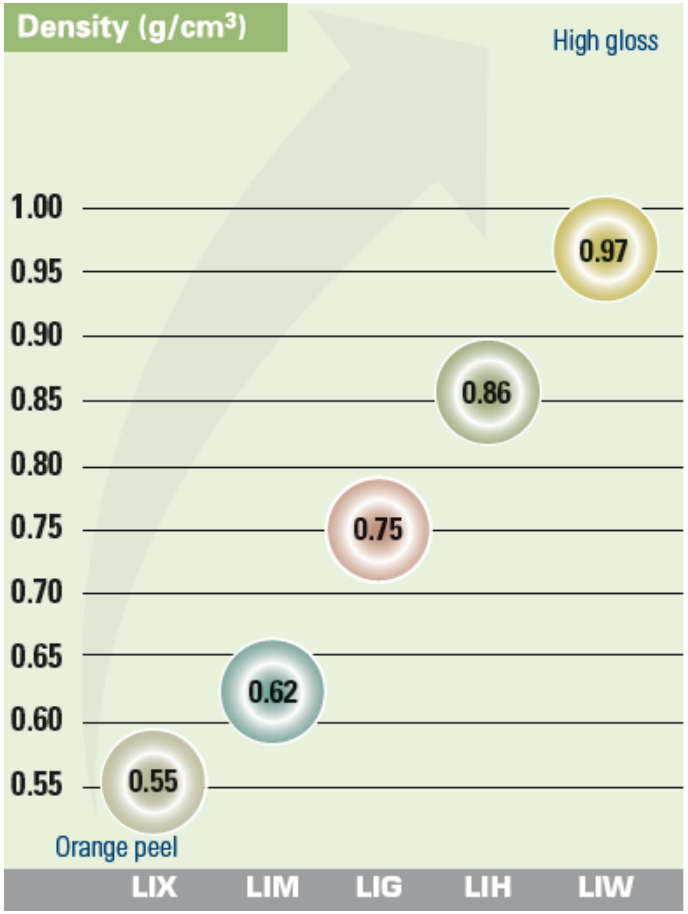
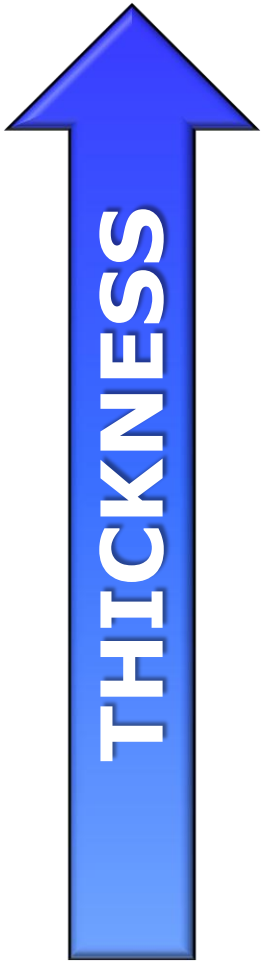
## Label Stiffness = density + thickness




**IML films (µm)**

70
65
60
55
50
45
(38)

**ALL Printing Technologies**

**Roll to Roll Technology**

# Criteria: container size

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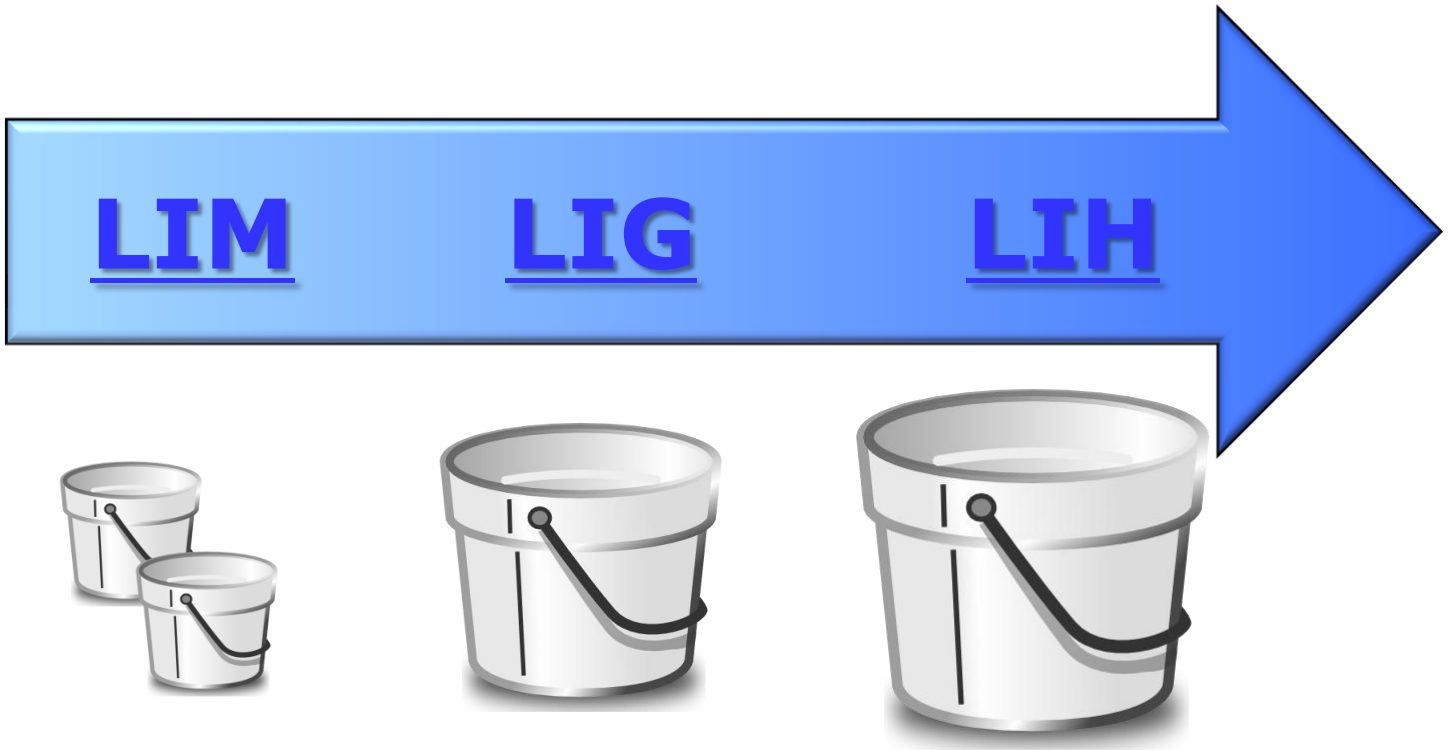
## DISTORTION/WRINKLE FREE CONTAINER



# Thermoforming-IML: LFE

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## LFE

- **Orange peel**
- High yield
- New thickness: 50 µm

| Nominal Thickness (µm) | Unit Weight (g/m²) | Yield (m²/kg) | After injection moulding |
|---|---|---|---|
| 65 | 40.3 | 24.8 | Orange Peel |
| | | | |
| | | | |
| | | | |
| Density (g/cm³): 0.62 | | | |

# New developments in IML

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- high performance films
- thin wall smooth finish
- barrier
- metallised IML
- digital printing

# LIU -Next generation IML

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BEST IN CLASS

# LIU

| µm | 50 | 60 |
|---|---|---|
| g/m² | 27.5 | 33.0 |

- Ultra high yield
- Reduced thickness : **50 µm**
- **High** post moulding **orange peel**
- **SUPERIOR** converting performance

# White voided : LIH

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LIH

- **Glossy** post moulding finish
- New thickness: 50 µm

**Thickness**: 50 µm
**Unit weight**: 43.0 g/m$^2$
**Density** : 0.86 g/m$^2$

- Thin wall applications

# Barrier IML

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**Goal:** EXTEND PRODUCT SHELF LIFE

EXTENDO® BARRIER TECHNOLOGIES **High barrier films (OTR < 3 $cm^3/m^2/d$)**

***Ti* solution:** LAMINATED labels

**EXTENDO® XTMU** (transparent) / LIM (white voided)

**Features and Benefits:**

- improved gas barrier (aroma barrier)
- mineral oil barrier
- superior scratch resistance (sandwich print)
- high gloss
- cost effective against monoweb cast PP

# Metallic IML

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Titanium™
METALLIC IML FILM

**Technical benefits:**

- Ease of handling (no secondary converting operations)
- For sheet fed and rotary printing
- Low curl
- No metal scuffing (protected metallic layer)
- Strong ink anchorage during print converting
- Designed for static charge label placement

**Targeting injection moulded packaging for:**

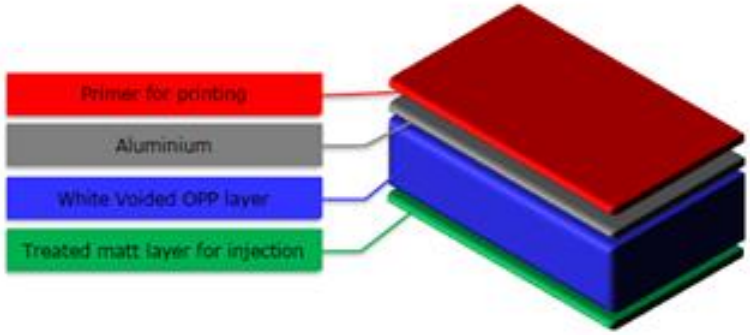
- **Industrial pail applications (e.g. paints & coatings)**
- **Neutraceutical products (e.g. sports supplements)**
- **Premium brand foodstuffs**

# Metallic IML

## D568

Description: metallized "LIH" with primer

**Thickness**: 71 µm
**Unit weight**: 60.7 g/m$^2$
**Density** : 0.86 g/m$^2$
**OD**: 2.5




**Features and Benefits:**

- Improved printability and metal adhesion
- No static pinning (alternative to paper)
- Good yield

- **Cost effective**

# Metallic IML : barrier

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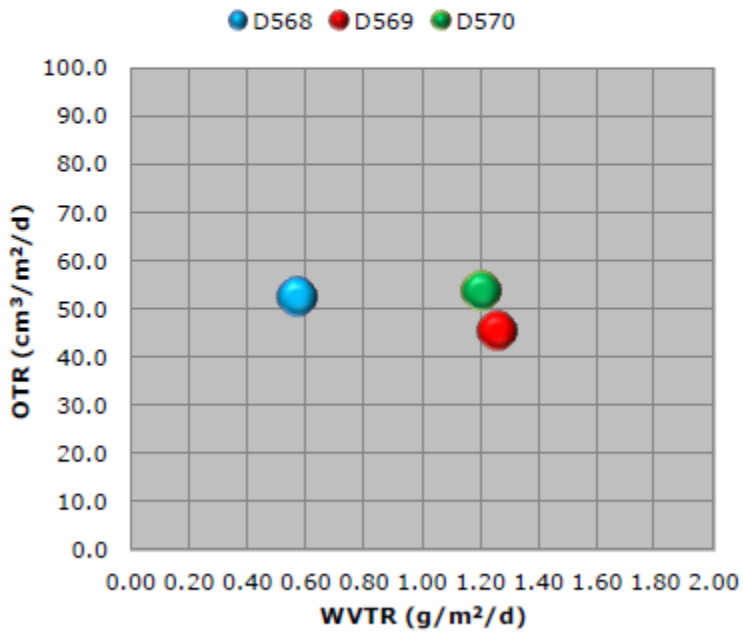
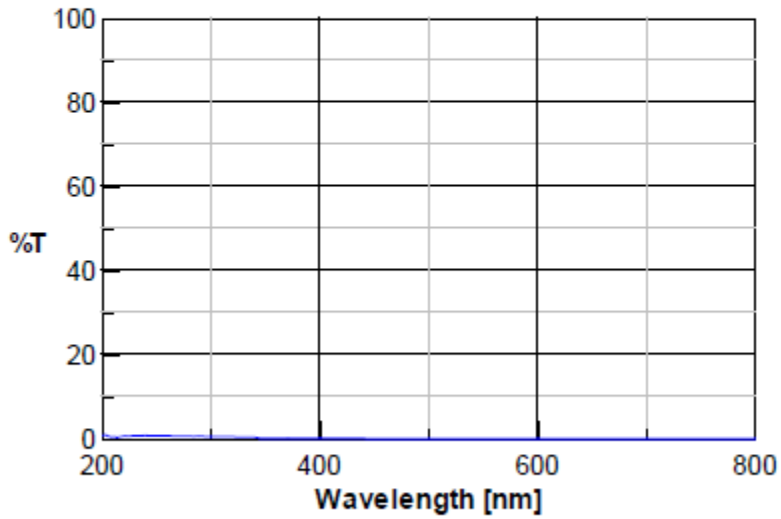
## Barrier Properties

*WVTR + OTR*




*UV and VISIBLE protection*




Average %T (200-400 nm) ≤ 1.0 %

# IML films: light barrier

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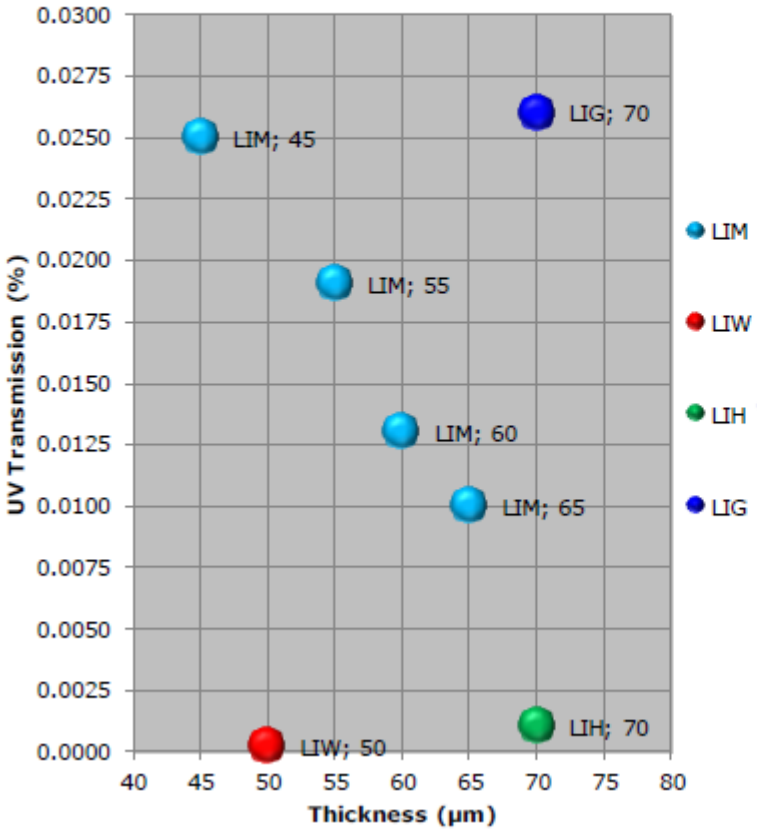
## IML Labels
NON – Trasparent materials:




Thanks to [$TiO_2$]

**UV PROTECTION**

*No need of further barrier*

# Digital printing for IML

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**Liquid toner**

- Digital print share: 80%
- HP
- Primer needed
- Preliminary tests: DONE

**Dry toner**

- Digital print share: 20%
- Xeikon
- Primer needed for IML
- LIT, LIH, LIG qualified

**Ink-jet**

- Emerging
- Higher speeds